

then $\frac{1}{2x} + x = y$. By Quadratic formula, this gives

$$x = \frac{y \pm \sqrt{y^2 - 4}}{2} \quad \text{If } y = -1 \text{ then } \sqrt{(-1)^2 - 4} = \sqrt{-3} \notin \mathbb{R}$$

so there is no real x that satisfies $f(x) = -1$.

7.2) $\forall a, b, c \in \mathbb{Z}^+ [(c|ab \wedge \gcd(a, c) = \gcd(b, c) = 1) \rightarrow c = 1]$

This shows only $c=1$ can be relatively prime to both a & b when $c|ab$. Otherwise by Theorem 7.2.2, it can be relatively prime to one.

Proof: By Theorem 7.2.2, $c|a$ & $c|b$

$$\text{so } \exists l \in \mathbb{Z} (cl = a) \text{ \& } \exists m \in \mathbb{Z} (cm = b)$$

Also by Theorem 7.1.4, we choose s_1, t_1, s_2 & t_2 so that

$$s_1 a + t_1 c = 1 = \gcd(a, c) \quad \&$$

$$s_2 b + t_2 c = 1 = \gcd(b, c)$$

$$\text{so } s_1 a + t_1 c = s_1 cl + t_1 c = c(s_1 l + t_1) = 1 \text{ so } c|1$$

$$\text{so } c \leq 1. \text{ since } c \in \mathbb{Z}^+, c = 1.$$

$$\text{If } a \in \mathbb{Z}^+, b \in \mathbb{Z}^+ \& a|b \quad \longrightarrow \quad a \leq b$$

Proof: supp $a > b$. we consider 2 cases. since $a|b$,

$$\text{so } \exists k \in \mathbb{Z} (ak = b) \quad \text{If } k = 0 \text{ then } b = 0. \text{ so we consider 2}$$

cases only after assuming $a > b$

Case 1) $k > 0$, then as $a > b$ so $ak > b$ which contradicts $ak = b$.

Case 2) $k < 0$. Then $b = ak < 0$ which contradicts $a, b \in \mathbb{Z}^+$

so $a \leq b$. so in previous result, $c=1$ follows as $c|1$ & $c \in \mathbb{Z}^+ \& 1 \in \mathbb{Z}^+$.

This result is true for all numbers so its definitely true for primes as well.

It is possible that a number a that is not prime, satisfies

$a|bc$ but $a \nmid b$ & $a \nmid c$. Ex: $4|6 \times 6$ but $4 \nmid 6$.

But for primes, By Theorem 7.2.3, if $p|ab$, it must divide atleast one of a or b (or both).

7.2) 4) $\min(e, f) + \max(e, f) = e + f$.

Case 1) $e = f$. Then clearly goal follows

Case 2) $e > f$. Then $\min(e, f) + \max(e, f) = f + e = e + f$.

Case 3) similar to case 2.

7.2) 6) $\rightarrow$ supp $\gcd(a, b) = 1$. As $\gcd(a, b)$ can be written as $sa + tb$
so $\exists s, t \in \mathbb{Z}$, $sa + tb = \gcd(a, b) = 1$

$\leftarrow$ supp $sa + tb = 1$. We will prove by contradiction. supp $d = \gcd(a, b) > 1$. (As $\gcd(a, b) = d$) By exercise 5, this means

a & b have a common prime factor p . So $p|a$ & $p|b$

$\exists k \in \mathbb{Z}^+ (pk = a)$ & $\exists l \in \mathbb{Z}^+ (pl = b)$

$\rightarrow$ this is either d or is in prime factorization of d

so $sa + tb = pk + pl = 1 \rightarrow p(k + l) = 1$ so $p|1$

but $p > 1$ so this is contradiction.

7.2) 7) supp $a, b \in \mathbb{Z}^+$ & $a', b' \in \mathbb{Z}^+$, $\gcd(a, b) = 1$, $a'|a$, $b'|b$.

If a' & b' have common ^{prime} p such that $p|a'$ & $p|b'$ then $p|a$ & $p|b$ so $\gcd(a, b) = p > 1$ which is not possible. So a' & b' have no common prime factor. By exercise 5, a' & b' are Relatively Prime.

7.2) 9) $d = \gcd(a, b)$, $d|a$, $d|b$

$a = jd$, $b = kd$

$\rightarrow \gcd(j, k) = 1$

supp j & k have a common prime factor. Let it be p .

Then $p|j$ & $p|k$. (As $p \nmid j$) so $pl = j$ & $pn = k$. So

$a = pld$ & $b = pnd$. We know there are $s, t \in \mathbb{Z}$, $d \cdot sa + tb$. So $d = spld + tpnd$. So $d = pd(sl + tn)$

so $pd \mid d$ but this is a contradiction as p is prime so

$p > 1$ so $pd > d$ so $pd \nmid d$.

so j & k have no common prime factor. By exercise 5,

$\gcd(j, k) = 1$.

7.2) 10) $a, b, d \in \mathbb{Z}^+$, $d \mid ab$ $\longrightarrow \exists d_1, d_2 \in \mathbb{Z}^+$ ($d = d_1 d_2$ & $d_1 \mid a$ & $d_2 \mid b$)

$dk = ab$ — (1)

let $d_2 = \gcd(a, d)$ so $d_1 \mid a$ & $d_1 \mid d$.

$d_1 j = a$ — (2) $d_1 d = d$ — (3)

let $l = d_2$. Then (3) $\rightarrow d_1 d_2 = d$. Also $d_1 \mid a$. To see that $d_2 \mid b$:

(1) $\rightarrow dk = ab \iff d_1 k = d_1 j b \iff k = j b$ so $j \mid k$. $l \mid j b$

But note by exercise 9, j & l are relatively prime so by theorem 7.2.2, $l = d_2 \mid b$.

7.2) 11) Supp q & r are Quotient & remainder when $m = l$

so $m = ql + r$, $0 \leq r < l$. ($l = \text{lcm}(a, b)$)

$a \mid m \rightarrow ak = m$

$l = \text{lcm}(a, b) \mid m$

$b \mid m \rightarrow bj = m$

also $a \mid l$ & $b \mid l$ so $aq = l$ & $bh = l$

Now $m = ql + r \iff ak = qag + r \iff a(k - qg) = r$ — (1)

similarly $b(j - qh) = r$ — (2)

If $k - qg = 0$ or $k = qg$, then $a(0) = r$ so $r = 0$. We

are proof By Contradiction. Supp $k - qg \neq 0$.

Subtracting & rearranging (1) & (2), we get

$a(k - qg) = b(j - qh)$

As $k - qg \neq 0$ & $a \neq 0$ so $a(k - qg) = b(j - qh) \neq 0$ so $j - qh \neq 0$

this concludes from (1) & (2) that $a \mid r$ & $b \mid r$ but since

It is $\text{lcm}(a, b)$ so $k \geq 1$ which is contradiction. Therefore $k \cdot q \cdot q = 0$, so $k = 0$ & hence $\text{lcm}(a, b) = m$ so $\ell \mid m$.

7.2) 12) By def of lcm , $m = \text{lcm}(a, b)$. To see $m = \text{lcm}(\text{lcm}(a, b), c)$. We will prove $\text{lcm}(a, b) \mid m$ & $c \mid m$ & m is smallest such number. The former follows from exercise 11 & latter is assumption. Now we prove

$$\forall m' \in \mathbb{N} \left[(\text{lcm}(a, b) \mid m' \wedge c \mid m') \rightarrow m' \geq m \right]$$

$$\text{supp } m' \in \mathbb{N}, \text{lcm}(a, b) \mid m' \text{ \& } c \mid m' \rightarrow m' \geq m.$$

As $a \mid \text{lcm}(a, b)$ so $a \mid m'$. Similarly $b \mid m'$ but m is lcm of a & b so $m' \geq m$.

7.2) 15) (a) sup $x \in H$ & $y \in H$ so $x = 4n+1$ & $y = 4m+1$
 $\rightarrow xy = (4n+1)(4m+1) = 16mn + 4n + 4m + 1 = 4(4mn + n + m) + 1 \in H$.

(b) We will use strong induction. sup $k \in \mathbb{N}$ &

$$\forall k \in \mathbb{N} [k > 1 \rightarrow (k \text{ is Hilbert prime or product of Hilbert primes})]$$

We will prove this for $\mathbb{N}$. If $\mathbb{N}$ is Hilbert prime, then there is nothing more to prove. So sup $\mathbb{N}$ is not a Hilbert prime, then $\mathbb{N} = xy$ where $x \in H$ & $y \in H$. Also

$1 < x < \mathbb{N}$ & $1 < y < \mathbb{N}$ so by inductive hypothesis, x & y are either Hilbert primes or product of Hilbert primes.

In all cases, $\mathbb{N}$ is product of Hilbert primes.

(c) $441 = 9 \times 49$

Also $441 = 21 \times 21$

7.2) 17) $a, b, c \in \mathbb{Z}^+$

(a) $\text{gcd}(a, bc) \mid \text{gcd}(a, b) \cdot \text{gcd}(a, c)$

let $d_1 = \text{gcd}(a, bc)$, $d_2 = \text{gcd}(a, b)$ & $d_3 = \text{gcd}(a, c)$

so $d_1 \mid a$ & $d_1 \mid bc$ so d_1 divides any linear combination.

of a & bc . We can write $d_2 d_3$ as linear combination of a & bc .

($d_2 d_3$) By theorem 7.1.4, let

$$d_2 = s_2 a + t_2 b \quad \& \quad d_3 = s_3 a + t_3 c$$

$$\text{so } d_2 d_3 = s_2 s_3 a^2 + s_2 t_3 a c + s_3 t_2 a b + t_2 t_3 b c$$

$$= (s_2 s_3 a + s_2 t_3 c + s_3 t_2 b) a + (t_2 t_3) b c$$

so $d_1 | d_2 d_3$.

(b) We know that $\forall a \forall b \forall c \in \mathbb{N} (\gcd(a, b) | \gcd(a, bc))$.

Because $\gcd(a, b) | b$ & $b | bc$ so $\gcd(a, b) | bc$.

similarly $\gcd(a, c) | \gcd(a, bc)$ so by exercise 11,

$$\text{lcm}(\gcd(a, b), \gcd(a, c)) | \gcd(a, bc)$$

(c) Note that $\forall a \forall b \forall c \in \mathbb{N} [\text{lcm}(a, b) | c \rightarrow (a | c \wedge b | c)]$

This is because $a | \text{lcm}(a, b)$ & $\text{lcm}(a, b) | c$ ($\therefore$ By Assumption)

so $a | c$. similarly $b | c$.

Note that it's not necessary that $ab | \text{lcm}(a, b)$

$$\gcd(b, c) = 1 \quad \longrightarrow \quad \gcd(a, bc) = \gcd(a, b) \cdot \gcd(a, c)$$

As $\gcd(a, b) | b$ & $\gcd(a, c) | c$ & b & c are relatively prime.

so by exercise 7, $\gcd(a, b)$ & $\gcd(a, c)$ are relatively prime.

Using formula for lcm,

$$\text{lcm}(\gcd(a, b), \gcd(a, c)) = \frac{\gcd(a, b) \times \gcd(a, c)}{\gcd(\gcd(a, b), \gcd(a, c))}$$

$$= \gcd(a, b) \cdot \gcd(a, c)$$

$\therefore$ Denominator = 1

By part (b), $\gcd(a, b) \cdot \gcd(a, c) | \gcd(a, bc)$ & combining

this with part (a) completes the proof.

If a & b are relatively prime, then $\text{lcm}(a, b) = ab$

7.2) 20) let $a = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$, $b = q_1^{b_1} q_2^{b_2} \dots q_m^{b_m}$

so $a^2 = p_1^{2e_1} p_2^{2e_2} \dots p_k^{2e_k}$, $b^2 = q_1^{2b_1} q_2^{2b_2} \dots q_m^{2b_m}$

Since a^2 & b^2 are both even (as shown in theorem 6.4.5),

their prime factorizations contain 2. so

$a^2 = 2^{2e_1} p_2^{2e_2} \dots p_k^{2e_k}$, $b^2 = 2^{2b_1} q_2^{2b_2} \dots q_m^{2b_m}$

Since $a^2 = b^2$ so $a^2 = 2^{2e_1} p_2^{2e_2} \dots p_k^{2e_k} = 2^{2b_1+1} q_2^{2b_2} \dots q_m^{2b_m} = b^2$

By uniqueness of prime factorization, it follows that

$2^{2e_1} = 2^{2b_1+1}$ so $2e_1 = 2b_1+1$ but $2e_1$ is even

& $2b_1+1$ is odd so $2e_1 = 2b_1+1$ is a contradiction.